

Title Page: Generation, Survival, and Evolution of Open Complex Giant Systems: Inference Deductions and Isomorphic Proof of OCGS Based on the Non-IID First Principle

Target Journal: *IEEE Transactions on Systems, Man, and Cybernetics: Systems* (or *IEEE Transactions on Cybernetics*)

Author & Institutional Information

- **Full Name:** Fanchun Meng (Grit Meng)
 - **Current Position:** Former Lead Architect and Head of Global Supply Chain Integrated Planning Solution (IPS) System, Lenovo Group | Creator of Intelligent Planning and Control (IPC) Engine | Independent Researcher
 - **Author Affiliation:** Value Chain Physics & Complexity Systems Research Lab, Lenovo Group Global Supply Chain Alumni Network
 - **ORCID Identifier:** <https://orcid.org/0009-0003-8603-119X>
 - **Corresponding E-mail:** gritmeng@outlook.com
 - **SSRN Working Papers:** 7451920, 7442100, 7251098
 - **Open-Source Codebase & Repository:** <https://github.com/GritMeng/Value-Chain-Physics>
-

Manuscript Metadata

- **Document Type:** Regular Original Research Paper
 - **Running Head:** Non-IID First Principle & Universal OCGS Isomorphic Proof
 - **Estimated Word Count:** ~8,500 words
 - **Keywords:** System and Complexity Science; Open Complex Giant Systems (OCGS); Non-Independent and Non-Identically Distributed (Non-IID); Vector Cancellation; Metasynthetic Wisdom; Department of Overall Design; Qian's Isomorphism Proof; Banach Fixed Point Theorem
-

Author Declarations & Funding

- **Funding Statement:** This research received no specific grant from funding agencies in the public, commercial, or not-for-profit sectors. All algorithm development and computational experiments were executed using independent private computing infrastructure.

- **Conflicts of Interest:** The author declares no financial, personal, or professional conflicts of interest that could influence or bias the scientific conclusions presented in this work.
- **Data & Code Availability:** All core topological data structures, algorithm specifications, and verification benchmarks supporting the conclusions of this paper are available in the open-source repository at [GitHub: Value-Chain-Physics](<https://github.com/GritMeng/Value-Chain-Physics>).